

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer specializing in building scalable microservices and REST APIs. Proven expertise in Java, Spring Boot, Spark, and cloud platforms like AWS and Azure, with a Master of Science in Computer Science. Adept at CI/CD automation, database optimization, and delivering robust, high-availability solutions, including asset management and full-stack development.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, managing over 50,000 daily API requests with 99.99 percent SLA uptime.
- Automated CI/CD pipelines with Jenkins and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling to reduce issue resolution time by 20 percent.
- Led an EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated in Agile/Scrum teams to deliver FHIR-compliant healthcare solutions, employing Git, SVN, and comprehensive JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed Java and EJB-based enterprise applications and RESTful web services for automotive clients, ensuring robust full-stack capabilities.
- Optimized system performance by 90 percent through meticulous SQL query tuning, database optimization, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab, integrating automated testing and quality gates for enhanced code reliability.
- Mentored junior developers through code reviews, debugging sessions, and advocating for clean coding practices in complex projects.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization strategies.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity in asset management contexts.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical application modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, ensuring timely project milestones.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process over 8 GB of NYC taxi trip data in near real-time, supporting advanced analytics.
- Engineered an interactive Flask web application for data exploration with dynamic filtering and visualization capabilities.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services, including S3, RDS, ECS, Lambda, and QuickSight for asset management.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, demonstrating advanced AI capabilities.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, enhancing secure AI development.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment of AI models.
- Focused on developing innovative solutions within the domain of AI and data privacy.

SKILLS

- Programming Languages: Proficient in Java and Python for building robust applications and data pipelines; extensive experience with SQL for complex data manipulation and database operations.
- Frameworks and Libraries: Expertise in Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, and LoRA for developing scalable REST APIs and AI solutions; skilled in JUnit for comprehensive unit testing.

- Web Services and Full-Stack: Advanced capabilities in designing and implementing REST APIs and SOAP APIs; experience in full-stack development for end-to-end solution delivery.
- Cloud Platforms: Skilled in AWS (EC2, RDS, S3, Lambda) and Azure (Functions, DevOps, ServiceBus) for scalable infrastructure and application deployment; familiar with Google Cloud Platform.
- Databases and Tools: Experienced with PostgreSQL, Oracle Database, IBM DB2, and RDS; proficient in Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for efficient development and infrastructure automation.
- Concepts and Methodologies: Deep understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/ CD, Microservices, large- scale data platforms, infrastructure automation, and privacy-preserving LLM-based applications; experience with blockchain principles and asset management strategies.